

Hamza Aziz

AI Engineer | Generative AI & Computer Vision

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PROFESSIONAL SUMMARY

AI Engineer with 3+ years of full-time experience building production voice AI, LLM applications and computer-vision systems. Helped lift live ordering success on a real-time, multi-tenant Google Gemini DriveThru voice-ordering platform from ~25–30% to 60%+ across 4 POS integrations; previously built face-recognition attendance across 20+ live CCTV streams at >99.99% accuracy in controlled UAT, and worked in a team building a voice ERP assistant.

SKILLS

- **Generative & Conversational AI:** Google Gemini, Vertex AI, Context Caching, LangGraph, LangChain, Agentic AI, RAG, LLMs, vLLM, Prompt Engineering, Rasa, NLP
- **Machine Learning & Computer Vision:** PyTorch, TensorFlow, Scikit-learn, OpenCV, YOLO, InsightFace, NumPy, Pandas
- **Inference & Edge AI:** TensorRT, ONNX, DeepStream, NVIDIA Jetson, DGX A100
- **Programming & Backend:** Python, SQL, Bash, TypeScript, FastAPI, Socket.IO
- **Data, Messaging & DevOps:** PostgreSQL, MongoDB, Redis, Milvus (Vector DB), Kafka, MinIO, Docker, CI/CD, Git, Linux

WORK EXPERIENCE

AI Product Engineer *VoicePlug Inc.* | September 2025 – Present

- Helped raise live client success metrics from ~25–30% to 60%+ through prompt and conversation-flow fixes addressing intent, entity, menu-understanding and fallback failures; converted insights from 500+ real customer interactions into improvements with product, support and QA teams
- Develop and operate a production FastAPI and Socket.IO voice-ordering platform deployed at 2 live restaurants serving 4+ QSR brands; implemented Spanish-language ordering mode
- Optimised conditional Flash-Lite entity extraction before one cached Gemini 2.5 Flash agent call with deterministic guards and no model tool calls; implemented per-store Vertex AI context caching, Flash/Flash-Lite tiering across five model roles, a 1,024-token reasoning cap and per-call token-cost accounting, holding a 2.63 s median reply over 17,884 production turns
- Maintain a per-session Gemini cart worker with schema-bound updates and Redis locking, where checkout drains queued jobs before application code handles provider-specific pricing and finalisation; fixed cart queue and locking issues and refined entity parsing, size normalisation and combo/multi-size guards to prevent cart crashes and invalid writes
- Implemented fuzzy matching for ASR disfluencies, LLM-generated fillers, delayed fillers for stalled turns, a no-LLM pickup fast path and duplicate-TTS suppression
- Maintain 4 POS integrations (QUBeyond, Clover, Speedline, Deliverect) with provider-specific pricing and commit flows, Jinja2 prompts, Redis session state and a MongoDB order book
- Built replay tests checking spoken replies and cart writes, running each case 3 times for model variability, and an ops review loop linking production logs, review sheets and replay to prompt/code fixes; improved exception handling and fallback escalation, reducing recurring production issues by 6%

Assistant Engineer – AI *Global InfoVentures Pvt. Ltd.* | August 2023 – August 2025

Facial-Recognition Attendance Systems (CCTV, PTZ and edge)

- Designed and built the CCTV attendance application from scratch as sole developer, integrating InsightFace-REST (SCRFD, w600k_r50) with Milvus and achieving >99.99% accuracy against the attendance register in controlled UAT at the main and library gates; delivered 4 CCTV, PTZ and edge variants using TensorRT and DeepStream across DGX A100, RTX 4090 and Jetson Orin Nano

- Processed 20+ live CCTV streams with ROI masking and low-light gating, batching 32 frames across 4 recognition threads on nginx-balanced TensorRT containers on A100 MIG slices; per-camera Kafka events feed Raspberry Pi ID-and-face displays
- Automated face-crop collection and nightly staging-gallery updates from elevated-angle footage, improving real-world recognition accuracy from 85% to 99%; ported models using ONNX and combined mobile/CCTV datasets, improving cross-domain recognition by 14%
- Built FastAPI reporting and date/location-partitioned PostgreSQL ingestion, with ERP synchronisation scheduled at 1-minute intervals, acknowledgement tracking and manual recovery for a 20-camera main-gate deployment with average daily footfall of ~2,000 people
- Built and deployed a PTZ classroom-attendance system as sole developer: ONVIF preset sweeps and RTSP capture with faculty captures at lecture start and end, InsightFace-REST inference with Milvus matching and YOLOv9 headcounts on a GPU MIG slice; stored lecture-wise attendance in year-partitioned PostgreSQL tables, with ERP exports scheduled at 5-minute intervals, success flags and manual resend

G6 Voice ERP Assistant

- Developed G6: a Dockerized FastAPI voice ERP assistant configured for 4 institutes, with Google speech recognition, gTTS replies and two-stage Rasa routing for student, employee and fee queries; achieved near-100% intent classification accuracy using 50+ voice query variations per module, with student/employee lookup disambiguation and clarification handling
- Implemented schema-grounded query generation using 100+ field-level schema mappings per module and per-institute Milvus retrieval of aliases, values and few-shot examples, reducing LLM hallucinations and improving SQL query precision by over 30%; Gemini emits fields and conditions rather than full SQL, and application code applies aliases, filters and joins to ERP payloads; vector-based field/value disambiguation improved SQL generation precision by over 40%

AI Intern *Global InfoVentures Pvt. Ltd.* | February 2023 – July 2023

- Delivered 10+ AI/ML training sessions, mentored 50+ students across CSE, AIML and Data Science, and guided 10+ YOLOv8/CNN object-detection projects through deployment
- Built a browser-based PPE violation monitor using a custom YOLOv7 model, Flask, OpenCV and PyTorch, with a 0.75 confidence gate and Gmail API email alerts with cooldown control

EDUCATION

Birla Institute of Technology and Science (BITS), Pilani – WILP 2025 – 2027
Master of Technology (Executive) — in progress

- Artificial Intelligence and Machine Learning

Dr. A. P. J. Abdul Kalam Technical University, ABES Institute of Technology
Bachelor of Technology 2019 – 2023

- Computer Science and Engineering

CERTIFICATES

- Rasa Developer Certification — 2025
- Introduction to Data Science, IBM — 2025
- LangChain — Develop AI Agents with LangChain & LangGraph, Udemy — 2025
- Generative AI with Large Language Models, Coursera — 2024
- Building Video AI Applications at the Edge on Jetson Nano, NVIDIA Deep Learning Institute — 2024
- Getting Started with AI on Jetson Nano, NVIDIA Deep Learning Institute — 2024
- Artificial Intelligence and Machine Learning, Global InfoVentures Pvt. Ltd. — 2023
- Getting Started with Deep Learning, NVIDIA Deep Learning Institute — 2022